Task 1: Guided Activity, User profile with missing data

```
import Foundation
struct Address {
   var city: String?
}
struct User {
   var name: String
   var age: Int?
   var address: Address?
}

func describe(_ user: User) -> String {
   guard let age = user.age else {
      return "\(user.name): Age is unknown"
   }
   let city = user.address?.city ?? "unknown city"
   return "\(user.name), Age: \(age), City: \(city)"
}
let user1 = User(
   name: "Anu",
   age: 21,
   address: Address(city: "Chennai")
)
let user2 = User(
   name: "Ben",
   age: nil,
   address: Address(city: "Coimbatore")
)
let user3 = User(
   name: "Chitra",
   age: 22,
   address: Address(city: nil)
)
print(describe(user1))
print(describe(user2))
print(describe(user3))
let number1 = Int("42")
let number2 = Int("abc")
print("Valid conversion:", number1 as Any)
print("Invalid conversion:", number2 as Any)
```

OUTPUT:

```
Anu, Age: 21, City: Chennai
Ben: Age is unknown
Chitra, Age: 22, City: unknown city
Valid conversion: Optional(42)
Invalid conversion: nil
```

Task 2: Practice Task, Safe number parsing

```
import Foundation
let inputs = ["10", "x", "25", "", "7.5", "40"]

let validNumbers = inputs.compactMap { Int($0) }
print("Valid integers:", validNumbers)

let sum = validNumbers.reduce(0, +)
print("Sum:", sum)

let invalidCount = inputs.count - validNumbers.count
print("Invalid entries:", invalidCount)

let firstDoubled = validNumbers.first.map { $0 * 2 }
print("First valid number doubled:", firstDoubled as Any)
```

OUTPUT:

```
Valid integers: [10, 25, 40]
Sum: 75
Invalid entries: 3
First valid number doubled: Optional(20)
```

Task 3: Debug Task, Remove the crashes.

```
import Foundation
let scores = ["Asha": 90, "Ravi": 72]
if let kiran = scores["Kiran"] {
   print("Kiran's score:", kiran)
} else {
   print("Error: Kiran's score was not found.")
}

if let number = Int("12a") {
   print("Number:", number)
} else {
   print("Error: '12a' is not a valid integer.")
}

if let kiran = scores["Kiran"], let number = Int("12a") {
```

```
    print("Sum:", kiran + number)
} else {
    print("Cannot calculate sum because one or both values are missing or invalid.")
}
```

OUTPUT:

```
ERROR!
Error: Kiran's score was not found.
Error: '12a' is not a valid integer.
Cannot calculate sum because one or both values are missing or invalid.
```